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An Information Retrieval System for the Cranfield Dataset

GitHub Link: https://github.com/poorvijn/ir_system

Dataset: https://ir-datasets.com/cranfield.html

- Developed at Cranfield University UK for evaluating IR systems
- Documents: 1,400 documents from aeronautical engineering domain
- Each document contains: Title, abstract/text content
- Queries: 225 natural language queries
- Relevance Judgments (Qrels)
    - Human labeled relevance data
    - Maps query to the relevant document IDs
    - Enables evaluation using metrics like Precision and Recall
- Domain-specific to aerodynamics and fluid mechanics

Features:

1. Data Loading
    - Loaded dataset using ir_datasets library
    - Extracted:
        - document IDs and text
        - queries
        - relevance judgments

2. Text Preprocessing
    - Converted text to lowercase
    - Removed special characters using regex
    - Tokenized text into words

3. TF-IDF Indexing
    - Used vector space model
    - Converted documents into numerical vectors using TF-IDF
    - Captures importance of terms relative to corpus

4. Inverted Representation
    - Achieved via TfidfVectorizer

- Maps terms to document weights

5. Query Processing
   - User queries processed using same preprocessing
   - Converted into TF-IDF vector

6. Document Ranking
   - Used cosine similarity
   - Compared query vector with document vectors
   - Returned top-k most relevant documents

7. Evaluation (Offline)
   - Computed Precision@5
   - Compared retrieved documents with ground-truth relevance (qrels)
   - Averaged over multiple queries

8. Interactive CLI Interface
   - User can:
     - Enter custom queries
     - View ranked results
   - Displays:
     - document ID
     - similarity score
     - text snippet

Functions used:

1.Libraries
- ir_datasets: Load dataset (docs, queries, qrels)
- scikit-learn: TfidfVectorizer for TF-IDF computation and cosine_similarity for ranking
- Re: Text cleaning and removing punctuation

2.Custom Functions
- In utils.py
  - preprocess(text)
    - Lowercases text
    - Removes special characters
    - Tokenizes into words
- In indexer.py
  - build_index(documents)
    - Creates TF-IDF matrix from documents
  - get_vectors()
    - Returns document vectors

    - get_vectorizer()
        - Returns fitted TF-IDF model
    - get_doc_ids()
        - Returns list of document IDs
- In retriever.py
    - search(query, top_k)
        - Converts query to vector
        - Computes cosine similarity
        - Returns top-k ranked documents
- In evaluator.py
    - precision_at_k(results, relevant_docs, k)
        - Computes precision for top-k results
- In main.py
    - load_nfcorpus(max_docs)
        - Loads Cranfield dataset (docs, queries, qrels)
    - main()
        - Controls full pipeline:
            - Loading
            - Indexing
            - Evaluation
            - CLI interaction

Sample Input/Output:

```
(venv) pro@DESKTOP-DHOAKOA:/mnt/c/users/pro/desktop/ir_system$ python3 main.py
Loading dataset...
Building index...
/mnt/c/users/pro/desktop/ir_system/venv/lib/python3.12/site-packages/sklearn/feature_extraction/text.py:526: UserWarning: The parameter 'token_pattern' will
 not be used since 'tokenizer' is not None'
  warnings.warn(

Running queries...

Query: what similarity laws must be obeyed when constructing aeroelastic models
of heated high speed aircraft .
Top results: ['13', '51', '184', '12', '486']
Precision@5: 1.00

Query: what are the structural and aeroelastic problems associated with flight
of high speed aircraft .
Top results: ['12', '51', '746', '792', '884']
Precision@5: 0.60

Query: what problems of heat conduction in composite slabs have been solved so
far .
Top results: ['5', '485', '181', '399', '144']
Precision@5: 1.00

Query: can a criterion be developed to show empirically the validity of flow
solutions for chemically reacting gas mixtures based on the simplifying
assumption of instantaneous local chemical equilibrium .
Top results: ['166', '1275', '1189', '1312', '1085']
Precision@5: 0.20

Query: what chemical kinetic system is applicable to hypersonic aerodynamic
problems .
Top results: ['103', '746', '943', '1032', '1374']
Precision@5: 0.00

Query: what theoretical and experimental guides do we have as to turbulent
couette flow behaviour .
Top results: ['491', '386', '472', '257', '775']
Precision@5: 0.40

Query: is it possible to relate the available pressure distributions for an
```

```
IR System Ready
Type your query or 'exit' to quit


Enter query: are real-gas transport properties for air available over a wide range of
enthalpies and densities .
Top Results:
1. DocID: 302 | Score: 0.2792
   approximations for the thermodynamic and transport properties of high temperature air . the thermodynamic and transport properties of high-temperature...
2. DocID: 524 | Score: 0.2281
   stagnation point heat transfer in partially ionized air .   comparison of heat-transfer rates, obtained by using transport properties recently reporte...
3. DocID: 1199 | Score: 0.2193
   theoretical investigations of a supersonic laminar boundary layer with foreign-gas injection .   the phenomena arising from the uniform injection of h...
4. DocID: 493 | Score: 0.2148
   real-gas laminar boundary layer skin friction and heat transfer .   the laminar-boundary-layer equations have been integrated for the case of a flat p...
5. DocID: 1214 | Score: 0.2109
   the drag of elongated bodies over a wide reynolds number range .   the resistance of bodies in motion through an incompressible viscous fluid is predi...

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